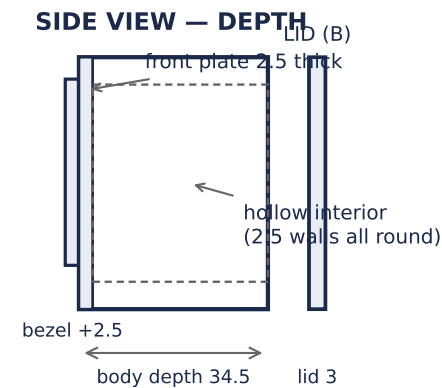
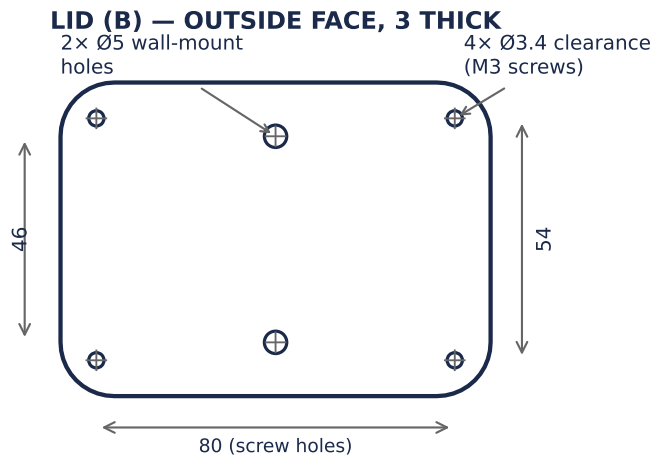
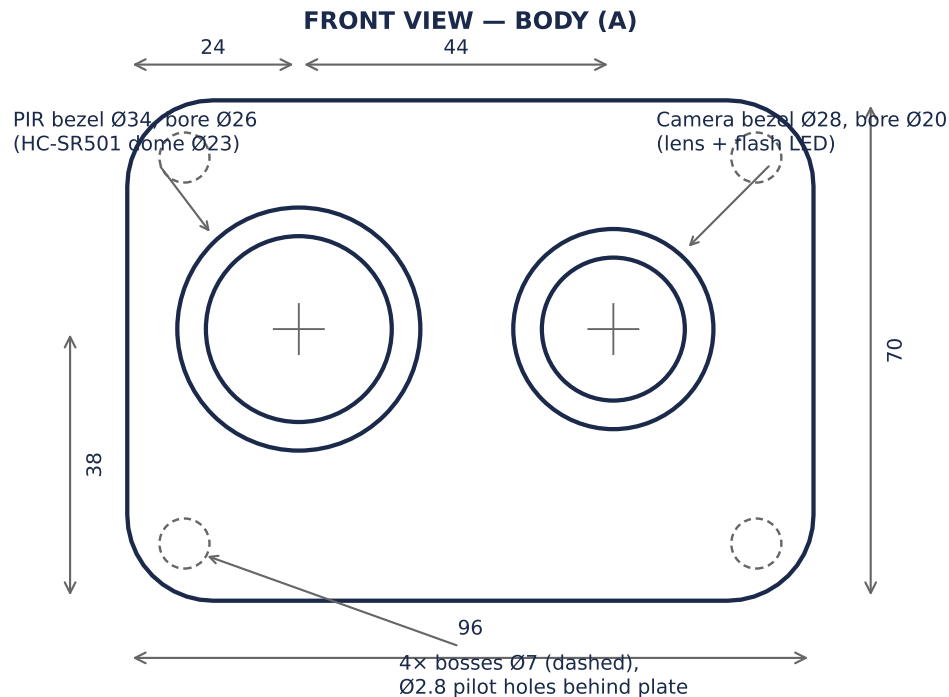


WhatsApp Doorbell Cam — Pebble Enclosure

Dimensioned views · Body (A) and Lid (B) · STLs in the case folder

Rev 2026-10-02

Units: mm · JakinsCraft



BOTTOM VIEW — USB NOTCH



cable notch 14 wide x 5.5 tall (x 41-55), behind the front plate

PRINTING

- PETG recommended (porch sun/heat). 0.2 mm layers, 3 walls, 15% infill.
- Body prints FRONT-FACE-DOWN; lid prints flat. No supports needed.
- Both STLs are single watertight solids – they open directly in Bambu Studio (fixed 2026-10-02; the old files were not).

BEFORE A LONG PRINT

- Test-fit your parts: HC-SR501 dome is Ø23 (bore Ø26); ESP32-CAM boards vary slightly between sellers. Print, check, then commit.
- USB power cable routes out through the bottom notch.

Assembly — WhatsApp Doorbell Cam

Exploded stack, parts list, wiring and build order

Rev 2026-10-02

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PARTS LIST (* FTDI is for flashing only)

#	Part	Qty
A	Enclosure body (3D print, PETG)	1
B	Enclosure lid (3D print, PETG)	1
1	ESP32-CAM (AI-Thinker) + OV2640	1
2	HC-SR501 PIR motion sensor	1
3	M3 × 8 self-tapping screws	4
4	470 µF capacitor (across 5V/GND)	1
5	USB 5V wall adapter + cable	1
6	FTDI USB-to-TTL programmer *	1

ASSEMBLY ORDER

1. Print A and B. Dry-fit the PIR dome and camera board.
2. Flash the ESP32-CAM with the FTDI before it goes in.
3. Seat the PIR behind the Ø26 bore, dome poking through.
4. Seat the ESP32-CAM behind the Ø20 bore, lens + flash out.
Fix boards with hot glue or foam tape.
5. Wire per the table below; put the 470 µF capacitor across 5V and GND, close to the ESP32-CAM.
6. Route the USB cable through the bottom notch.
7. Fix the lid (B) to the wall: two screws through the Ø5 holes.
8. Close the body onto the lid with the 4 M3 screws.

WIRING

From	To	Note
PIR VCC	5V (USB)	PIR runs on 5V
PIR GND	GND	common ground
PIR OUT	ESP32-CAM GPIO 13	wake-up trigger
USB 5V +	ESP32-CAM 5V pin	wall adapter power
USB GND	ESP32-CAM GND	—
470 µF cap	across 5V / GND	stops brown-outs

